A Market World Model for Public LLMs: Complete, Honest, Auditable Crypto Cognition before Decision

Anonymous Author(s)

Abstract

Most quant stacks start from *strategy*. Public LLMs (GPT-, DeepSeek-, GLM-, and Gemini-class agents) that act in cryptocurrency markets fail earlier: they lack a complete, honest, auditable *market cognition base*. We present an anonymized *financial world-model runtime* that compiles asynchronous multi-band evidence into a point-in-time (PIT) world bundle. Epistemic observations $O_{j,t} = (x, \tau, q, g, r)$ and compilation Π_t yield $\mathcal{F}_t^{\mathrm{AI}}$; completeness, honesty (B, U, H), and WMI/ACWMI support a typed abstention contract; evidence IDs make actions auditable. Four live public LLMs (temperature 0) on the *same* 100 OOS asset-days per arm show a three-arm behavior ladder: Compiled thin-world abstain 1.0; Ungated (same content, no hard flag) mean 0.68 (range 0.43–0.86); Raw 0.04–0.75. Disclosure helps but is vendor-dependent; the typed boolean enforces full cross-vendor refusal, and models that act under Ungated lose money on thin support. A non-trading grounding task shows the same asymmetry for analysis: with the compiled bundle, models identify the missing evidence bands near-perfectly (F1 0.96–1.00); from a raw feed, F1 collapses to 0.00–0.29. Economic CE is a secondary nonempty-content probe—not a strategy Holy Grail. We detail the runtime architecture (collectors, vintage store, BandPIT clock, bundle schema, LLM adapter) as a systems contribution for ICAIF.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; *Machine learning*; • **Applied computing** → *Economics*.

Keywords

world models, public LLMs, AI agents, cryptocurrency, point-in-time, abstention, trustworthy AI, market cognition, systems

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1 Introduction

Public LLMs are increasingly asked to analyze or trade crypto markets. The bottleneck is not another alpha signal. It is *understanding*: before decision, an agent needs a market world that is **complete** (missingness disclosed), **honest** (stale or role-incoherent evidence gated), and **auditable** (actions bound to evidence). Venue fragmentation, derivatives, on-chain flows, news, and macro vintages arrive on incompatible clocks [11, 13]. Generative world models [5, 6, 9] and tool-using finance agents [18–20] typically assume a usable observation stream. Crypto agents often do not have one.

Thesis. We build a market *world-model runtime* for public LLMs: compile raw multi-band evidence into a PIT-safe cognition bundle that GPT / DeepSeek / GLM / Gemini-class models can call. Strategy metrics are secondary probes that the world has content—not the product objective. In slogan form: *quant from strategy; this runtime from understanding.*

What we claim (and do not). RQ1 tests whether a typed world interface can *enforce* world-conditional abstention across live public LLMs—a runtime contract, not a claim that models spontaneously discover thinness from raw ticks. RQ2 tests whether compiled band fields are economically nonempty versus same-input momentum. We do not claim unconditional trading alpha or a Dreamer-style generative simulator.

Contributions.

(1) **Cognition semantics.** Epistemic observations, compilation Π_t, a lag reconstruction bound, completeness/honesty fields, WMI/ACWMI abstention, evidence-bound actions.
(2) **Anonymized runtime architecture.** Vintage-aware collectors, BandPIT previous-close clock, quality-tagged bundles, availability shocks O_t, OpenAI-compatible multi-vendor adapter, and a frozen Compiled / Ungated / Raw consumer protocol.
(3) **Live three-arm validation.** Four live public LLMs on identical 100-day samples: Compiled enforces thin-world abstain 1.0; Ungated soft judgment averages 0.68; Raw often trades. A grounding task shows compiled worlds make epistemic answers verifiable (missing-band F1 0.98 vs. 0.12 raw); a secondary content probe shows vintaged band fields are economically nonempty.

2 Related Work

World models in AI.. World models learn compact states and often dynamics [5, 6, 9]. We address the complementary *observation-side* problem: a PIT-safe, quality-tagged crypto world *before* dynamics or LLM policy. We claim the missing systems layer generative WMs usually assume—not a Dreamer/JEPA simulator.

LLM agents in finance. ChatGPT return prediction [12], open financial LLMs [17], memory-augmented trading agents [10, 19], and tool-augmented multimodal agents [20] advance agent skill. Agentic tool use [18] still assumes aligned inputs. We evaluate within-model Compiled vs. Ungated vs. Raw under a frozen schema so the estimand is the *world interface*, not a new memory module or prompt trick.

Selective prediction and trustworthy AI.. Abstention can be optimal under noise [2, 3]. We tie refusal to world quality via an explicit runtime contract, not classifier confidence alone.

PIT measurement and crypto microstructure. Macro vintages [1] and crypto microstructure [11, 13] motivate multi-band compilation.

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Bootstrap discipline for secondary probes follows [7, 15, 16]. Asset-pricing ML [4, 8, 14] is adjacent: we validate nonempty compiled content for agents, not priced factors.

3 Market World-Model Semantics

Definition 1 (Epistemic observation). *An epistemic observation is*

$$O_{j,t} = (x_{j,t},\ \tau_{j,t},\ q_{j,t},\ g_{j,t},\ r_{j,t}):$$

value, latest-available time, quality, main-view gate, semantic role. A bare feature dump that omits (τ, q, g, r) *is not a world observation.*

Definition 2 (Financial world state). $W_t = \Pi_t(\mathcal{F}_t^{\mathrm{raw}})$ *with scalar quality* $q(W_t)$ *(e.g. WMI/ACWMI) gating abstention;* $\mathcal{F}_t^{\mathrm{AI}} = \sigma(W_t)$.

Definition 3 (Compilation). $\Pi_t = B_t \circ M_t \circ A_t$*: align clocks, apply honesty/missingness gates, assemble band views.*

Completeness, honesty, quality. Ready/limited/missing counts disclose coverage. $\mathrm{WMI}_t = B_t U_t H_t$; ACWMI supports regime-conditional gating. The Compiled contract exposes the abstain flag (`should_ai_abstain`) and `thin_world` as first-class fields.

Assumption 1 (Bounded lag and increments). *Band* j *has max observation lag* Δ_j*; latent state increments satisfy* $\|S_u - S_{u-1}\| \le \bar{\delta}$.

Proposition 1 (Lag reconstruction bound). *Under Assumption 1,*

$$\|\widetilde{S}_t - S_t\| \ \le\ C_{\mathrm{del}}\bar{\delta} + C_{\mathrm{noi}}\varepsilon_t + C_{\mathrm{miss}}m_t.$$

Compilation cannot erase delay; it can refuse to pretend the bound is zero.

Intuition. Crypto agents that concatenate tool dumps silently absorb delay and missingness into the feature vector. Prop. 1 makes those terms explicit: a thicker raw payload can worsen the reconstruction if honesty collapses.

Proposition 2 (Compilation ≠ feature dump). *Enlarging raw span without* Π_t *need not enlarge usable* $\mathcal{F}^{\mathrm{AI}}$*: ungated stale evidence can raise volume while lowering honesty/stability.*

Proposition 3 (World-conditional abstention). *If non-abstain actions exceed a world-dependent cost* $c_{\mathrm{abs}}(W_t)$*, the Bayes action is abstain; with costs decreasing in WMI, abstention forms a lower set in world quality [2].*

Proposition 4 (LOBO content vs. gating). *Deleting band* j *changes value through a content channel and a gating channel. Empirically we report content share under an ungated rule so the probe speaks to nonempty fields.*

Reading for systems reviewers. Props. 2–3 justify why the runtime exports gates and abstain flags rather than only tilts. Prop. 4 justifies the secondary economic probe: if deleting a durable band does not change actions, the bundle is decorative.

Remark 1 (Not a generative WM). *We do not learn* $p(s_{t+1} \mid s_t, a_t)$*. We deliver a* state compiler + abstention runtime.

Table 1: Anonymized module map (systems view).

Module	Responsibility
Collectors	Ingest multi-band series with timestamps / vintages
Vintage stores	Persist raw, market, and analytics objects
BandPIT compiler	Previous-close readiness, Π_t, WMI/ACWMI
Bundle builder	Assemble complete/honest/auditable JSON world state
Shock index O_t	Query availability outages for quasi-exogenous levers
LLM adapter	OpenAI-compatible chat; frozen prompts; temp. 0
Eval harness	Compiled/Ungated/Raw sampling, transcripts, tables

4 Runtime Architecture

We describe the anonymized prototype as a layered systems object (Figure 1). Source paths and product names are omitted for double-blind review; an anonymized artifact will be released upon acceptance.

4.1 Layered design

(1) **Data layer (collectors).** Exchange, macro, alternative, and (when available) news/on-chain/options collectors write vintage-aware history: observation timestamps plus macro `available_at` vintages.
(2) **Store layer.** Embedded analytical stores hold raw series, merged market panels, and pipeline outputs (readiness, WMI/ACWMI, paper world snapshots).
(3) **Logic layer (compilation).** BandPIT reconstructs band readiness on a previous-close clock; honesty/missingness gates and band assembly implement Π_t; world-quality indices and availability shocks O_t are first-class.
(4) **Service layer.** A read API serves quality-tagged AI bundles and health/readiness objects to consumers without requiring collector restarts for newly arrived rows.
(5) **Consumer layer.** Frozen-prompt LLM / rule agents call an OpenAI-compatible adapter (temperature 0) under Compiled, Ungated, or Raw treatments.

4.2 Data flow: inputs, processing, outputs

Table 2 traces the runtime end-to-end. *Inputs*: per-band collectors pull public market data—exchange klines/funding/open-interest via REST, daily macro risk indices (equity volatility, dollar index), aggregate stablecoin circulation, and (when reachable) news/on-chain/options feeds—each row stored with its observation timestamp and, for macro, an `available_at` vintage. *Processing*: for every asset-day, BandPIT selects the latest observation with timestamp $\le (t{-}1)$ 23:59, grades it fresh/stale/missing against per-band age thresholds, aggregates breadth/stability/honesty into WMI and regime-conditional ACWMI, derives content tilts (5-day macro-index changes; 7-day stablecoin-flow slope; return momentum), and logs availability shocks O_t. *Outputs*: a ~3,990-row PIT panel (10 assets × 399 days, ~50 columns), per-day world bundles (Listing 1), and the consumer transcripts/tables behind Sections 5.

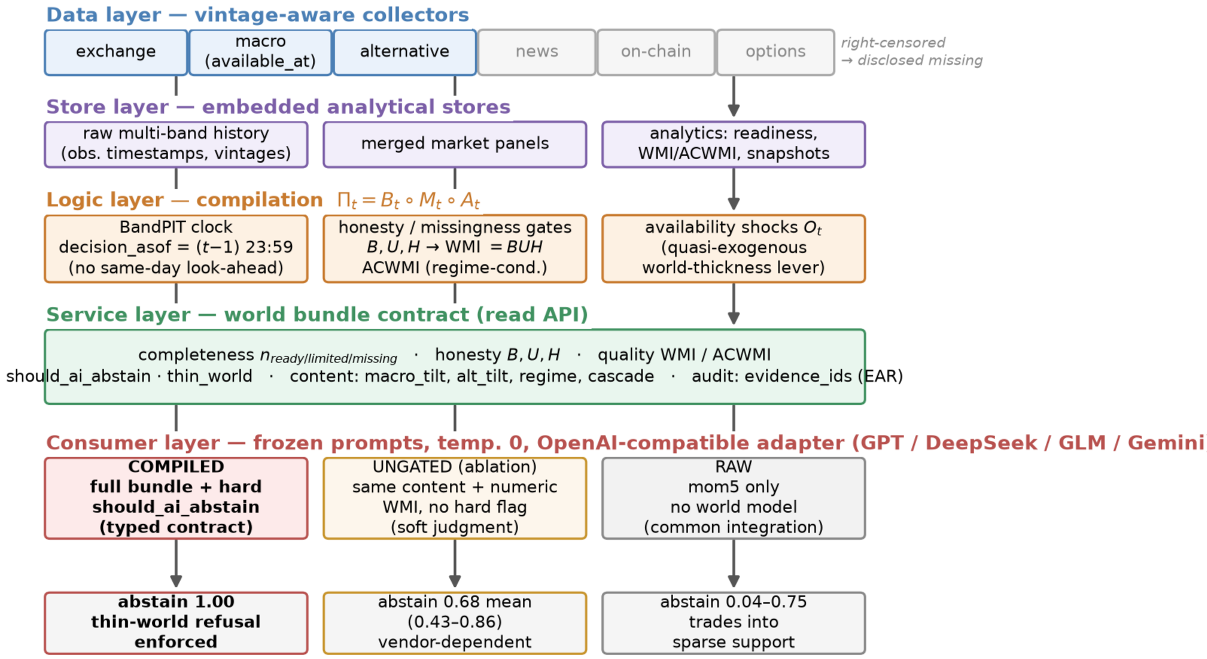

Figure 1: Anonymized runtime architecture. Five layers map raw multi-band evidence to a typed world-bundle contract; the consumer layer runs the frozen three-arm protocol (Compiled / Ungated / Raw), whose live abstention outcomes are shown at the bottom.

Table 2: Data flow through the runtime (anonymized sources).

Band	Fetched (raw)	Emitted (bundle)
exchange	OHLCV klines (1h/1d), funding, open interest, orderbook	mom5; band status/age; funding context
macro	equity-vol and dollar indices; vintaged macro series	macro_tilt (5d changes); status/age
alternative	aggregate stablecoin circulation	alt_tilt (7d flow slope); status/age
news / on-chain / options / tokenomics	RSS, chain explorers, option chains (right-censored here)	status = `missing`, disclosed
payoff	daily closes (external reference)	close-to-close r_t for scoring only
derived	—	B, U, H, WMI, ACWMI, `should_ai_abstain`, `thin_world`, O_t, `evidence_ids`

4.3 PIT previous-close clock

For calendar day t, the decision information set is reconstructed at $(t-1)$ 23:59. The payoff is the same-calendar-day close-to-close return r_t. This removes same-day look-ahead in band statuses and vintaged macro/alternative content. The bundle records the clock in `decision_asof`, and every live run pins `timing_protocol` to the previous-close protocol.

Table 3: World bundle field groups (cognition base).

Group	Fields
Timing	`decision_asof`, previous-close protocol
Completeness	$n_{\text{ready/limited/missing}}$, band statuses
Honesty	B, U, H, main-view / stale gates
Quality	WMI, ACWMI, `should_ai_abstain`, `thin_world`
Content	macro_tilt, alt_tilt, regime, cascade
Audit	`evidence_ids`, EAR required

4.4 World bundle contract

Table 3 lists field groups. Listing 1 shows a compact Compiled example from the OOS panel: only 2/8 bands ready, WMI low, `should_ai_abstain=true`, and `evidence_ids` binding actions to disclosed bands/tilts. Ungated removes the hard boolean (and threshold) but keeps numeric WMI/ACWMI and completeness. Raw retains only `mom5` (plus a non-informative noise bit).

Listing 1: Compact Compiled bundle example (OOS thin day).

```
{"date":"2026-05-11","asset":"ADA",
 "decision_asof":"2026-05-10T23:59:00",
 "completeness":{"n_ready":2,"n_missing":6,
   "ready_share":0.25},
 "world_model_index":{"wmi":0.015,"acwmi":0.30,
```

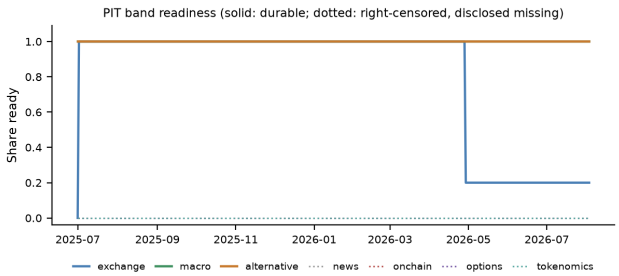

Figure 2: PIT readiness: missing bands are first-class in the bundle.

```
  "should_ai_abstain":true,"thin_world":true},
"macro_tilt":1.0,"alt_tilt":-1.0,
"detected_regime":"crisis",
"audit":{"evidence_ids":[
  "band:exchange:missing","band:macro:ready",
  "band:alternative:ready","tilt:macro:1.0"]}}
```

4.5 Three information arms and LLM adapter

- **Compiled (product contract):** full bundle; frozen prompt *requires* abstain when `should_ai_abstain` is true.
- **Ungated (ablation):** same content/completeness/numeric WMI; *no* hard flag; soft prompt asks the model to judge thinness.
- **Raw (without world model):** `mom5` only; no quality index or abstention guidance.

Actions ∈ {bullish, bearish, neutral, abstain}. Understanding metrics prioritize thin-world abstain rate and EAR proxy over CE. Durable archive bands: {exchange, macro, alternative}; news/onchain/options are right-censored and disclosed as missing (Figure 2). The adapter speaks OpenAI-compatible chat completions; model IDs are passed through; credentials stay in environment variables (never in artifacts).

4.6 Frozen prompt contracts

Listings 2–3 excerpt the frozen Compiled and Ungated contracts (temperature 0; action schema identical). The only intentional difference is hard vs. soft abstention. Raw prompts omit world-quality language entirely.

Listing 2: Compiled contract (excerpt).

```
Decision rules (excerpt):
1. If world_model_index.should_ai_abstain is true
   OR the world is thin, you MUST "abstain".
2. Prefer band content over raw momentum
   (macro_tilt, alt_tilt are PIT-safe).
3. Rationale must be consistent with evidence_ids.
```

Listing 3: Ungated soft contract (excerpt).

```
Decision guidance (soft):
1. Judge whether missingness / low WMI/ACWMI
   makes action unsafe; you MAY "abstain".
2. Prefer band content when you act.
NO hard abstain boolean; NO forced policy.
```

Not "just instruction following." Compiled abstention is partly prompt obedience—and that is the *product requirement*: production agents need an enforceable refuse interface when $q(W_t)$ is low (Prop. 3). Ungated measures soft judgment from disclosure alone; Raw measures the common "price signal only" integration. Reliable Compiled refusal is the SLA; spontaneous Ungated refusal is a bonus.

Table 4: Live three-arm ladder on identical 100 OOS asset-days (Wilson 95% CIs). ΔCE secondary.

Model	Thin C	Thin U [CI]	Abs. R [CI]	ΔCE
gpt-5.4-mini	1.00	0.63 [.53,.72]	0.04 [.02,.10]	+0.73
deepseek-v4-flash	1.00	0.81 [.72,.87]	0.75 [.66,.82]	+1.22
glm-5.2	1.00	0.86 [.78,.91]	0.11 [.06,.19]	+1.04
gemini-3.5-flash-lite	1.00	0.43 [.34,.53]	0.07 [.03,.14]	+1.30
Mean	1.00	0.68	—	+1.07

4.7 Service surface and reproducibility

The service layer exposes read-only objects used by the consumer harness: (i) quality-tagged world bundles; (ii) band readiness / WMI–ACWMI snapshots; (iii) availability-shock queries O_t; (iv) health metadata disclosing whether paper engines or production proxies hydrate ACWMI inputs. Collectors and the logic pipeline can run as supervised jobs; the API reads stores live, so newly compiled rows appear without process bounce. For review we freeze prompts, temperature, action schema, IS/OOS cut, and sampling seed; model IDs and an OpenAI-compatible base URL are configuration, not estimands. Credentials never enter committed artifacts. Upon acceptance we will release an anonymized repository with the consumer harness, frozen prompts, and table-reproduction scripts.

5 Experiments

5.1 Setup

PIT panel ~399 days (≈13 months), 10 liquid names, previous-close clock, chronological IS/OOS 200/200 days. Live RQ1: all three arms run on the *same* stratified 100 OOS asset-days (same sampling seed), temperature 0, frozen prompts, OpenAI-compatible gateway. Models: `gpt-5.4-mini`, `deepseek-v4-flash`, `glm-5.2`, `gemini-3.5-flash-lite`. Offline mocks reproduce the protocol on full OOS without keys. Scarce/thin labels bind on OOS by design: primary test is honesty under sparse support, not a dense-world trading horse-race.

5.2 RQ1 (primary): Three-arm live behavior

Table 4 and Figure 3 are the headline result; all arms share the same 100 OOS asset-days. Under Compiled, all four models abstain on every thin-world day (thin-abs. 1.0); rationales cite world fields on 93–99% of days. Under Ungated, mean thin-abs. falls to 0.68 (range 0.43–0.86): DeepSeek and GLM stay above 0.80, GPT drops to 0.63, and `gemini-3.5-flash-lite` falls to 0.43 and trades. When Ungated models do act, they lose (CE −1.04 to +0.02 across models)—soft judgment fails exactly where enforcement matters. Under Raw, abstain rates are 0.04–0.75 with substantial directional mass (Table 5). Mean ΔCE (Compiled−Raw) = +1.07 is reported last: Compiled CE is zero under full abstention, so the gap mainly reflects Raw overtrading—not a Holy Grail.

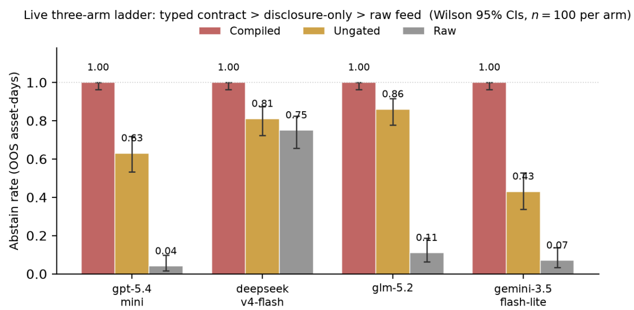

Figure 3: Live three-arm ladder (same 100 days per arm; Wilson 95% CIs). Typed contract > disclosure-only > raw feed; soft judgment is vendor-dependent.

Table 5: Raw-arm action mix (100 days). Compiled is 100% abstain.

Model	Abs.	Bull.	Bear.	Neut.
gpt-5.4-mini	0.04	0.28	0.38	0.30
deepseek-v4-flash	0.75	0.14	0.04	0.07
glm-5.2	0.11	0.40	0.48	0.01
gemini-3.5-flash-lite	0.07	0.44	0.49	0.00

Inside the Ungated arm. Ungated rationales cite disclosed world fields (missingness, readiness, WMI) on 53–93% of days, so the models demonstrably *read* the world; they differ in acting on it. Non-abstain Ungated actions are overwhelmingly directional (e.g. Gemini: 55 bearish, 2 bullish of 57 trades) and destroy value on thin support—evidence that vendor-specific caution, not lack of disclosure, drives the gap the typed contract closes.

Offline diagnostics. World-honoring offline followers abstain at 1.0 on full OOS; momentum-only and noisy mocks ignore thin-world flags (thin-abs. 0)—the Raw failure mode in miniature. Offline results are protocol checks; live vendors are the product claim.

5.3 RQ3: Non-trading cognition—world-state grounding

The cognition base should serve analysis, not only trade/abstain decisions. We add a live *grounding* task on 50 OOS asset-days per arm: the model must summarize the world state and answer (i) is the underlying data sufficient to act on, and (ii) *which evidence bands are missing*—a verifiable question with PIT ground truth. Table 6: with the Compiled bundle, all four models identify missing bands near-perfectly (F1 0.96–1.00); with Raw, F1 collapses to 0.00–0.29. Both arms mostly call the data insufficient (a bare momentum feed is obviously thin), so the discriminating skill is not caution but *verifiable epistemics*: only the compiled world lets the model say *what* is missing and be checked. This is the analyst-facing counterpart of Prop. 1's missingness term.

Table 6: Grounding task (50 days): can the model say *what is missing*?

	Missing-band F1		Insufficiency acc.	
Model	Comp.	Raw	Comp.	Raw
gpt-5.4-mini	0.96	0.29	0.96	0.86
deepseek-v4-flash	0.98	0.07	0.96	1.00
glm-5.2	0.98	0.11	0.98	1.00
gemini-3.5-flash-lite	1.00	0.00	1.00	0.94
Mean	0.98	0.12	0.98	0.95

Table 7: Secondary content probe (not the LLM product claim).

Policy / contrast	Sharpe	CE
Momentum always	0.101	−0.202
Thick ungated (content)	0.767	0.132
Mechanism − Momentum	–	0.334 (p=0.034)

5.4 RQ2: Nonempty compiled content

A transparent rule shares return inputs with momentum and adds vintaged macro_tilt / alt_tilt. Pre-specified OOS contrast mechanism − momentum: ΔCE=0.334 (p=0.034; stationary p=0.022). LOBO (Prop. 4): deleting macro or alternative collapses to momentum (content share 1.0 under the ungated rule). Relative gaps survive 10–25bps costs; absolute CE is cost-fragile. A 2017–2026 audit without vintaged archives finds no hidden return-rule alpha.

5.5 Discussion and threats

The live ladder $\mathrm{Abs}_C = 1.0 > \mathrm{Abs}_U \approx 0.68 > \mathrm{Abs}_R \in [0.04, 0.75]$ is the product evidence: a cognition base that marks the world thin induces vendor-stable refusal; disclosure alone is incomplete; raw feeds over-act. Cross-vendor heterogeneity under Ungated (GLM 0.86 vs. Gemini 0.43, non-overlapping Wilson CIs) is exactly why a typed contract is valuable in production: soft judgment is model-dependent, while Compiled refusal is stable. RQ2 shows band fields are not decorative. Claim: *understanding first, strategy second.*

Relative to FinMem / FinAgent-style systems [19, 20], we do not propose a new memory or tool router. We propose the observation compiler those agents implicitly need when the market feed is asynchronous and quality-heterogeneous. Relative to return-prediction LLM studies [12], we score refusal and auditability before PnL.

Threats to validity. (1) *Prompt contract*—Compiled refusal is partly instructed; Ungated is the soft contrast. (2) *Thin archive*—scarce rate ≈ 1 on OOS stresses honesty, not dense calibration. (3) *Baselines*—Raw matches common thin integrations; Ungated is the same-content control. (4) *Sample size*—100 live days per arm (Wilson CIs reported); harness supports full ~2000-row OOS sweeps. (5) *Vendor routing*—gateway and model IDs vary; prompts/schema are frozen for within-model contrasts. (6) *Task scope*—the trading action space is a coarse directional schema; the grounding task (Sec. 5.3) is a first non-trading probe, and richer analyst workflows remain open.

6 Design Rationale for ICAIF Systems Readers

Three implementation choices are deliberate.

Export gates, not only features. Silent feature stores hide staleness. By placing completeness, honesty, and `should_ai_abstain` in the bundle, the runtime makes refusal a first-class machine-readable event rather than a free-text apology.

Previous-close PIT over wall-clock streaming. Intraday streaming demos look impressive but confound look-ahead. The previous-close clock is conservative and auditable: every live decision cites a `decision_asof` stamp that a reviewer can recompute from the panel.

Three arms instead of one leaderboard. A single Compiled-vs-holdout Sharpe invites Holy-Grail readings. Compiled / Ungated / Raw separates (i) typed enforcement, (ii) soft judgment from disclosure, and (iii) no world model—matching how teams actually integrate LLMs.

7 Limitations

We do not claim generative dynamics or unconditional trading alpha. News/on-chain/options history is right-censored and disclosed as missing. Dense-world selective-prediction horse-races require thicker archives. Live samples are budget-constrained (100 asset-days per arm); the harness already supports full OOS replication. Camera-ready will restore author identity and point to an anonymized repository deferred during review.

8 Conclusion

Public LLMs need a market to *understand* before they decide. We contribute cognition semantics and an anonymized layered runtime—collectors, vintage store, BandPIT compilation, bundle contract, and LLM adapter—that turns asynchronous crypto evidence into a complete, honest, auditable world state. Live public LLMs exhibit a three-arm ladder: Compiled enforces thin-world refusal, Ungated partially recovers it from disclosure alone, and Raw often trades into sparse support. The same bundle makes analyst-facing answers verifiable (missing-band grounding F1 0.98 vs. 0.12 raw), and content probes confirm the compiled state is nonempty. Strategy remains downstream of understanding.

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